

$$19. \begin{cases} x_1 + hx_2 = 2 \\ 4x_1 + 8x_2 = K \end{cases}$$

The augmented matrix is: $\begin{bmatrix} 1 & h & 2 \\ 4 & 8 & K \end{bmatrix}$

Step 1: Row 2 = Row 2 - 4 × Row 1

$$\begin{bmatrix} 1 & h & 2 \\ 0 & 8-4h & K-8 \end{bmatrix}$$

The corresponding equations are:

$$\begin{aligned} 1. & x_1 + hx_2 = 2 \\ 2. & (8-4h)x_2 = K-8 \end{aligned}$$

a) By Theorem 2, the system has no solution if and only if $8-4h=0$ and $K-8 \neq 0$, i.e. $h=2$ and $K \neq 8$.

b) By Theorem 2, the system is consistent when at least one of the above conditions is False.

b) If $h \neq 2$, then by eqn 2, x_2 has a unique solution. Substituting value of x_2 in eqn 1, we get a unique solution of x_1 .

c) If $h=2$, $8-4h=0$. If $K=8$, $K-8=0$. In this case, eqn 2 reduces to $0=0$. Row 2 has no pivots, and hence x_2 is a free variable. By Theorem 2, the system has infinitely many solutions.

$$20. \begin{cases} x_1 + 3x_2 = 2 \\ 3x_1 + hx_2 = K \end{cases} \quad \text{The augmented matrix is: } \begin{bmatrix} 1 & 3 & 2 \\ 3 & h & K \end{bmatrix}$$

Step 1: Row 2 = Row 2 - 3 × Row 1

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & h-9 & K-6 \end{bmatrix}$$

The corresponding equations are:

$$\begin{aligned} 1. & x_1 + 3x_2 = 2 \\ 2. & (h-9)x_2 = (K-6) \end{aligned}$$

a) By Theorem 2, the system has no solution if and only if $h-9=0$ and $K-6 \neq 0$, i.e. $h=9$ and $K \neq 6$.

b) If $h \neq 9$, then in eqn 2, x_2 has a unique solution. By substituting value of x_2 in eqn 1, x_1 has a unique solution.

c) If $h=9$, $h-9=0$. If $K=6$, $K-6=0$. In this case, eqn 2 has been reduced to $0=0$. Row 2 has no pivots, and hence x_2 is a free variable. By Theorem 2, the system has infinitely many solutions.